Week 3: Updated Literature Review (Added Paper 12) *Jun 17 – 23*

Tyler, S., Olis, M., Aust, N., Patel, L., Simon, L., Triantafyllidis, C., Patel, V., Won Lee, D., Ginsberg, B., Ahmad, H., & Jacobs, R. J. (2025, September 24). Use of Artificial Intelligence in Triage in Hospital Emergency Departments: A Scoping Review. Dr. Kiran C. Patel College of Osteopathic Medicine (KPCOM). Cureus. https://doi.org/https://doi.org/10.7759/cureus.59906

Paper 1 – This scoping review examines how traditional emergency room triage (where staff manually sort patients into five urgency levels) often suffers from human error, high subjectivity, and emergency department overcrowding. To investigate this, the authors reviewed 29 primary research studies (in 2023) published over a 10 year period in major medical databases. The review found that many machine learning models achieved higher accuracy than traditional triage methods and were able to make predictions more quickly. However, a major downside noted by the authors is that these AI tools often assign patients to higher urgency categories than necessary, which can accidentally waste valuable hospital resources, and the models lack a clear way to explain why they made a decision which may make clinical adoption more difficult.

Da'Costa, A., Teke, J., Origbo, J. E., Osonuga, A., Egbon, E., & Olawade, D. B. (2025). AI-driven triage in emergency departments: A review of benefits, challenges, and future directions. *International Journal of Medical Informatics, 197, 105838*. https://doi.org/10.1016/j.ijmedinf.2025.105838

Paper 2 – This paper focuses on the clinical problem of emergency room overcrowding, uneven patient sorting during peak hours, and the heavy cognitive stress placed on hospital staff. The authors conducted a narrative review of peer-reviewed articles published over a 9 year period across databases like PubMed and Google Scholar. The reported outcomes show that AI tools were associated with shorter wait times, more consistent triage decisions and improved resource allocation (hospital beds, staffing). However, a major limitation the authors mentioned is that these systems can carry algorithmic biases that treat diverse patient populations unfairly, and there is currently an uncertainty around responsibility when AI systems make mistakes.

Araouchi, Z., & Adda, M. (2024). TriageIntelli: AI-Assisted Multimodal Triage System for Health Centers. *Procedia Computer Science, 251, 430–437*. https://doi.org/10.1016/j.procs.2024.11.130

Paper 3 – The authors investigated emergency room overcrowding and human judgement errors under the structured Korean Triage and Acuity Scale (KTAS). The authors trained six machine learning models on 1,267 emergency department records and used data-balancing techniques to address unequal class sizes. The reported outcomes show that an integrated "Stacking Model" achieved the highest classification accuracy of 80.05%, but the individual models had accuracies of roughly 78–80% standalone. A limitation identified by the authors is that the system relies on a single dataset source, which creates a need for further testing in other hospitals to ensure it functions reliably across different hospital networks.

Bhuvaneswari, E., Prasad, K. D. V., Ashraf, M., Jadhav, S., Rao, T. R. K., & Sudha Rani, T. (2025). A human-centered hybrid AI framework for optimizing emergency triage in resource-constrained settings. *Intelligence-Based Medicine, 12, 100311*. https://doi.org/10.1016/j.ibmed.2025.100311

Paper 4 – This study explores the issue of high human error rates, cognitive burnout and misallocated medical supplies in overcrowded emergency rooms operating under severe resource constraints. To solve this, the authors developed a hybrid AI prototype combining a machine learning model for risk prediction, an explanation feature showing why the model made a recommendation, and a resource management component that considered hospital bed and oxygen availability. The reported outcomes show that the system achieved a high 90% accuracy rate, reduced clinician decision-making time by 31% and improved the allocation of hospital resources by 20%. However, a key limitation stated by the authors is that the system was developed using a very small sample of real patient data, meaning it still requires testing in larger and more diverse hospital settings to determine whether it performs reliably.

Huang, J., Galal, G., Etemadi, M., & Vaidyanathan, M. (2022). Evaluation and Mitigation of Racial Bias in Clinical Machine Learning Models: Scoping Review. *JMIR Medical Informatics, 10(5), e36388*. https://doi.org/10.2196/36388

Paper 5 – This review examines racial bias in machine learning models, which can worsen existing healthcare inequalities if not addressed. To analyse this issue, the authors performed a systematic scoping review following standard PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-analyses) guidelines to screen 635 research papers, ultimately analysing 12 core studies that evaluated algorithmic fairness. The reported outcomes revealed that while 67% of clinical models showed evidence of racial bias, all reviewed studies that actively applied technical corrections successfully improved model fairness. However, a major limitation highlighted by the authors is that medical algorithms suffer from a severe lack of standardised reporting and missing demographic data, making it difficult for hospitals to reliably evaluate whether a tool is safe and unbiased before deploying it.

Agius, S., Cassar, V., Magri, C., Khan, W., Obe, D. A.-J., Caruana, G., & Topham, L. (2025). Predicting Emergency Severity Index (ESI) level, hospital admission, and admitting ward in an emergency department using data-driven machine learning. *BMC Medical Informatics and Decision Making, 25(1), 281*. https://doi.org/10.1186/s12911-025-02941-9

Paper 6 – This study looks at emergency department bottlenecks, unexpected patient surges, and the difficulty of accurately predicting hospital resource and bed needs at initial triage. To resolve this, the authors used the XGBoost machine learning algorithm to develop a two-stage model trained on a dataset of over 650,000 patient visits across 6 years. The model accurately predicted triage urgency, likelihood of admission and admitting ward, with performance improving when blood test results were included. However, a major stated limitation is that the system was developed using data from a single national hospital network, which creates a gap for external testing to ensure it works reliably in different regional or international health systems.

Almulihi, Q., Alquraini, A., Almulihi, F., Alzahid, A., Qahtani, S., Almulhim, M., Alqhtani, S., Alnafea, F., Mushni, S., Alaqil, N., Assiri, M., & Maghraby, N. (2024). Applications of Artificial Intelligence and Machine Learning in Emergency Medicine Triage—A Systematic Review. *Medical Archives, 78(3), 198*. https://doi.org/10.5455/msm.2024.236-239

Paper 7 – This systematic review evaluates severe emergency department overcrowding and the high potential for human error when clinicians attempt to sort complex, high-risk patient data under extreme time pressure. To investigate this, the authors executed a systematic review following standard PRISMA guidelines to filter and analyse recent peer-reviewed literature detailing machine learning tools in emergency triage. The review found that AI systems can improve the speed and accuracy of triage decisions when identifying high-risk patients. However, a major limitation stated by the authors is that most existing models suffer from limited dataset diversity and a lack of real-world clinical testing, which limits confidence in how well these systems would perform in different hospital settings.

Lee, S., Lee, J., Park, J., Park, J., Kim, D., Lee, J., & Oh, J. (2024). Deep learning-based natural language processing for detecting medical symptoms and histories in emergency patient triage. *The American Journal of Emergency Medicine, 77, 29–38*. https://doi.org/10.1016/j.ajem.2023.11.063

Paper 8 – This study investigates the issue of unused data in emergency departments where important clinical information is often recorded in free-text nursing notes that are difficult for traditional systems to analyse. The authors evaluated several natural language processing models designed to extract symptoms and medical history from triage notes. The best performing models achieved F1-scores above 88%, outperforming standard keyword based approaches. However, a major limitation stated by the authors is that the algorithm's performance depends heavily on the specific writing style and shorthand abbreviations of a single hospital, meaning it requires localised adjustment and retraining to deal with typographical errors or different medical slang elsewhere.

Tschoellitsch, T., Seidl, P., Böck, C., Maletzky, A., Moser, P., Thumfart, S., Giretzlehner, M., Hochreiter, S., & Meier, J. (2023). Using emergency department triage for machine learning-based admission and mortality prediction. *European Journal of Emergency Medicine, 30(6), 408–416*. https://doi.org/10.1097/MEJ.0000000000001068

Paper 9 – The authors examined the clinical problem of inefficient patient flow and unpredictable bed demands caused by traditional triage frameworks, which only measure immediate urgency rather than predicting downstream hospital admissions or patient mortality. To solve this, the authors used Random Forests and Neural Networks to analyse a dataset of 58,323 emergency patient records collected within ten minutes of arrival at a major Austrian hospital. The reported outcomes demonstrate that the machine learning models achieved high accuracy in predicting ward admissions and ICU placements, with age and chief complaint being important predictors. However, a key limitation identified by the authors is that the study was conducted in a single hospital facility and did not include physiological vital signs or blood test results, meaning the model requires further expansion and multi-centre testing to verify its safety elsewhere.

Gorick, H., McGee, M., Wilson, G., Williams, E., Patel, J., Zonato, A., Ayodele, W., Shams, S., Di Battista, L., & Smith, T. O. (2023). Understanding triage assessment of acuity by emergency nurses at initial adult patient presentation: A qualitative systematic review. *International Emergency Nursing, 71, 101334*. https://doi.org/10.1016/j.ienj.2023.101334

Paper 10 – In this review, the authors investigated the issue of workflow barriers and safety vulnerabilities in emergency departments, where rigid decision-making algorithms often fail to align with how triage nurses actually assess patient acuity under pressure. To investigate this gap, the authors conducted a qualitative systematic review of 28 primary studies using thematic synthesis and evaluated the quality of evidence. The reported outcomes reveal that nurses intentionally bypass algorithmic guidelines in favour of holistic reasoning and 'situational triage', altering their critical assessments to balance individual patient risks against external department pressures like overcrowding and low staffing. However, a major limitation stated by the authors is that their review had to exclude 12 potentially relevant full-text papers due to accessibility issues and restricted its sample exclusively to English-language publications.

Seo, Y. H., Lee, K., & Jang, K. (2024). Factors influencing the classification accuracy of triage nurses in emergency department: Analysis of triage nurses' characteristics. *BMC Nursing, 23(1), 764*. https://doi.org/10.1186/s12912-024-02334-9

Paper 11 – This study examines why patient sorting outcomes vary in emergency rooms, focusing on how nurse characteristics (personal traits) and the time they spend evaluating a patient affect sorting accuracy. To investigate this, researchers looked back at the medical records of 787 patients and 13 sorting nurses at a single hospital, using an expert panel to check if the initial urgency levels matched final patient outcomes. The results showed a baseline accuracy rate of 84.9%, revealing that nurses with more years of experience and those who took more time to evaluate patients were significantly more accurate. However, the authors note that their study is limited by its retrospective design and its focus on a single hospital, meaning more forward-looking research across different healthcare settings is needed to validate the findings.

De Freitas, L., Goodacre, S., O'Hara, R., Thokala, P., & Hariharan, S. (2020). Qualitative exploration of patient flow in a Caribbean emergency department. *BMJ Open, 10(12), e041422*. https://doi.org/10.1136/bmjopen-2020-041422 (Updated)

Paper 12 – This paper looks at how patients move through a large emergency department in Trinidad and Tobago to find out what speeds up or slows down their care. To do this, a researcher spent over 200 hours inside the hospital directly watching 143 patients from the moment they arrived until they left, while also talking with medical staff. The findings show that while smart moves like ordering basic medical tests early can save time, severe nurse shortages and a lack of open hospital beds cause massive delays. These bottlenecks disrupt the ED, forcing busy doctors to pause their own work to handle nursing duties or hunt down basic supplies. However, because this study only focused on a single hospital, more research is needed to see if the exact same problems happen in other emergency departments.

Updated Gap Analysis – Based on the literature review, the following gaps were identified. Gap 1 is selected as the primary focus for the proposed 12-week project.

Gap 1 (Primary Project Focus): Explainable AI + Clinician Trust
Evidence: Tyler et al. (2024) found that many AI triage models cannot explain the reason behind their recommendations. Da'Costa et al. (2025) identified clinician trust as a major barrier to adoption and emphasised the need for transparent systems. Bhuvaneswari et al. (2025) proposed a human-centred framework that incorporates explainable AI techniques, highlighting that accuracy alone may not be enough for clinical adoption. Lee et al. (2024) further showed that AI models can extract useful clinical information from nursing notes but their outputs remain difficult to verify without clear explanations. Gorick et al. (2023) found that triage nurses frequently rely on holistic reasoning and situational judgement rather than rigid algorithms, suggesting that decision-support systems should provide explanations that align with clinical reasoning processes.

What was missed: Many studies optimise for predictive accuracy, yet comparatively fewer focus on making recommendations understandable and verifiable to clinicians.

Pilot Idea: Develop an AI triage assistant that predicts a triage level and displays the factors influencing each recommendation.

Gap 2: Human-AI Collaboration
Evidence: Da'Costa et al. (2025) emphasised the importance of clinician trust, education and human oversight in AI-assisted triage. Bhuvaneswari et al. (2025) proposed a human-centred triage framework in which AI supports rather than replaces clinical decision-making. Almulihi et al. (2024) found that although AI systems often outperform traditional approaches, most studies lack real-world clinical deployment and evaluation. Seo et al. (2024) found that triage accuracy was significantly associated with nurse experience and evaluation time, suggesting that decision-support tools may be particularly valuable for supporting less experienced clinicians.

Additionally, De Freitas et al. (2020) demonstrated through qualitative process mapping that ED environments are highly unpredictable and resource-constrained, with severe staffing shortages forcing clinicians to constantly adapt through informal workarounds (such as doctors taking over nursing or porter tasks). This emphasises that successful implementation depends not just on the AI's math, but on how seamlessly the tool integrates into highly stressed, multitasking human workflows without adding cognitive debt. (Updated)

Together, these findings suggest that successful implementation depends not only on model performance but also on how clinicians interact with AI recommendations.

What was missed: Most studies evaluate whether AI performs better than clinicians, rather than whether clinicians perform better when supported by AI.

Pilot Idea: Evaluate whether AI decision support improves consistency among junior triage nurses when assigning patient acuity levels.

Gap 3 (Future Work): Performance Across Different Hospitals

Evidence: Araouchi & Adda (2024) reported that their highest performing triage model was developed using data from a single healthcare institution and explicitly identified the need for multi-centre validation. Da'Costa et al. (2025) highlighted concerns regarding algorithmic fairness and the need for testing across diverse patient populations. Agius et al. (2025) developed machine learning models for predicting Emergency Severity Index (ESI) level and hospital admissions but noted that validation was limited to a single emergency department. Similarly, Tschoellitsch et al. (2023) demonstrated strong performance in admission and mortality prediction using emergency triage data, yet their study was conducted in one Austrian hospital and lacked external validation. Lee et al. (2024) also found that natural language processing (NLP) models depend heavily on local documentation styles, suggesting that performance may vary when applied in different healthcare settings.

Crucially, De Freitas et al. (2020) emphasised that the majority of emergency care and patient flow research is conducted almost exclusively in developed countries, noting that their own single-site study in a developing Caribbean nation uncovered highly specific, localised bottlenecks. This uncovers a massive gap regarding whether predictive tools remain effective when transferred across entirely different socio-economic or regional healthcare infrastructure. (Updated)

What was missed: Many AI triage models are developed and tested using data from a single hospital, healthcare network or regional population, and mostly situated within well-resourced Western nations. As a result, it remains unclear whether these models would perform as well in different healthcare settings. (Updated)

Pilot Idea: Evaluate an existing triage model using a secondary dataset or simulated cross-site validation. This raises an important question: "Does an AI model trained in one setting remain reliable in another?"

Gap 1: Can clinicians understand the AI?
Gap 2: Can clinicians work effectively with the AI in a chaotic environment? (Updated)
Gap 3: Does the AI still work elsewhere (especially across different resource settings)? (Updated)

Literature Synthesis (Summary of the 12 Papers)

Across the reviewed literature, there is broad agreement that artificial intelligence can improve the speed, consistency and accuracy of emergency department triage. Multiple studies reported successful prediction of triage levels, hospital admissions, mortality risk and resource requirements using machine learning techniques. At the same time, recent research examining nursing practice highlights that triage decisions are influenced not only by clinical data but also by experience, situational judgement and environmental pressures within the emergency department. Observational research mapping clinical journeys emphasises that these environmental pressures frequently manifest as severe nursing shortages and resource scarcity, which force staff to constantly adapt through unpredictable, informal workarounds just to keep patients moving.

Several recurring challenges were identified across the literature. Many AI models operate as black boxes that provide little explanation for their recommendations, making them difficult for clinicians to understand and verify. Most systems have also been developed and validated using data from a single institution, raising concerns about generalisability and fairness across different patient populations. Crucially, the existing literature remains heavily skewed toward well-resourced healthcare systems in developed countries, leaving a major gap in understanding how these operational workflows and digital tools would function in developing regions like the Caribbean. Together, these studies suggest that future work should focus on developing transparent decision-support tools that complement clinical reasoning while remaining reliable across diverse healthcare settings.